PROIECT

Design Of Microstrip Patch Antenna At 2.4 Ghz Frequency

1. Introduction

A microstrip patch antenna is an internal antenna used at microwave frequencies, most microstrip antenna consist of multiple patches in a two dimensional array.A microstrip patch antenna is an internal antenna used at microwave frequencies, most microstrip antenna consist of multiple patches in a two dimensional array.A standard MPA features a radiating metallic patch placed over a dielectric substrate, with a ground plane beneath. The patch, often made from copper may assume different geometries such as rectangular, circular, or elliptical; however, the rectangular form is commonly preferred for its simplicity and ease of analysis. The patch antenna, microstrip transmission line and ground plane are made of high conductivity metal (typically copper). The patch is of length L, width W, and sitting on top the substrate (some dielectric circuit board) of thickness h with permittivity. Typically the height h is much smaller than the wavelength of operation, but should not be much smaller than 0.025 of a wavelength (1/40th of a wavelength) or the antenna efficiency will be degraded.

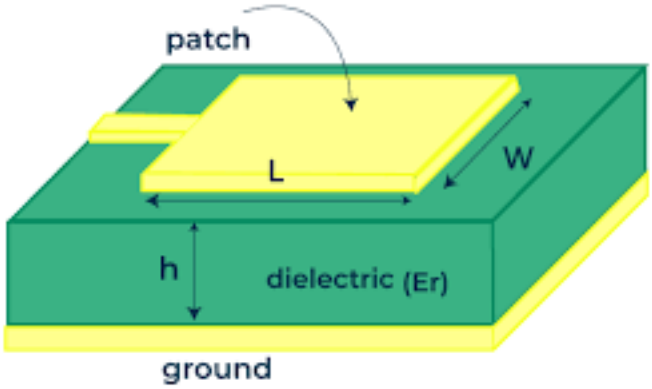

Fig 1

Working of *Microstrip* Patch Antenna

The microstrip patch contains a conducting patch on a dielectric slab whose other side is grounded. When current through a feed line reaches the strip present on the antenna,it leads to the generation of electromagnetic waves.The waves from patch generate a radiation pattern as waves begin to

radiate from the sides of the patch. The waves produced depends upon the thickness of the substrate, due to the small thickness of substrate , the waves get reflected from the edges.

Geometry of Microstrip Antenna

The frequency of operation of the patch antenna is determined by the length L. The center frequency is given by the formula:

$$f_c \approx \frac{c}{2L\sqrt{\varepsilon_r}}$$

We also have to keep in mind that the length of the antenna should be at least equal to one half of the wavelength within the substrate medium. We can also determine the wave number :

$$Wavenumber(k) = \frac{2\pi}{\lambda}$$

The width of the antenna controls the input impedance, the width can also increase the bandwidth of our antenna.Becouse of the size of the patch the input impedence is aproximately 250-300 ohm.The width also controls the radiation pattern for the coordinates of Phi and Theta for which we use the next formulas:

$$E_\theta = \frac{\sin\left(\frac{kW\sin\theta\sin\phi}{2}\right)}{\frac{kW\sin\theta\sin\phi}{2}}\cos\left(\frac{kL}{2}\sin\theta\cos\phi\right)\cos\phi$$

$$E_\varphi = -\frac{\sin\left(\frac{kW\sin\theta\sin\phi}{2}\right)}{\frac{kW\sin\theta\sin\phi}{2}}\cos\left(\frac{kL}{2}\sin\theta\cos\phi\right)\cos\theta\sin\phi$$

$$f(\theta,\phi) = \sqrt{E_\theta^2 + E_\phi^2}$$

We can also determine the length and the width so we can design our antenna in a very efficient way , also i have done the calculation of the effective dielectric constant that is based on the height, the dielectric constant ,and the width of the patch antenna.I have used this constant to calculate further the effective length of my antenna, and in the end the actual length of the patch:

$$W = \frac{c}{2f_o\sqrt{\frac{(\varepsilon_r + 1)}{2}}} \qquad \varepsilon_{eff} = \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2}\left[1 + 12\frac{h}{W}\right]^{-\frac{1}{2}}$$

$$L_{eff} = \frac{c}{2f_o\sqrt{\varepsilon_{eff}}} \qquad \Delta L = 0.412h\frac{(\varepsilon_{eff} + 0.3)\left(\frac{W}{h} + 0.264\right)}{(\varepsilon_{eff} - 0.258)\left(\frac{W}{h} + 0.8\right)} \qquad L = L_{eff} - 2\Delta L$$

The other parameter that I calculated is the length extension parameter that is a patch antenna calculation to compensate for fringing fields ensuring accurate resonance at the target frequency.Also I have used the notation εr for the relative permittivity of the substrate, and c is the speed of light that i used at the value of 3 x 10^8 meters /second.

2. *Simulations*

This is the model of my antenna, that I created, the ground plane and the patch are made of the material "Perfect E" which is the "perfect conductor" but in reality they are made out of copper , the patch is also containing the microstrip feedline that is designed with a 50 ohm characteristic impedance, featuring an excitation port that supplies the electromagnetic signal to the radiator. The substrate is also made of the material FR4-epoxy that has the height of 1.6mm and a ielectric constant of 4.4. I also created a "radiation box" to simulate the Far fields report , this box is made out of air.

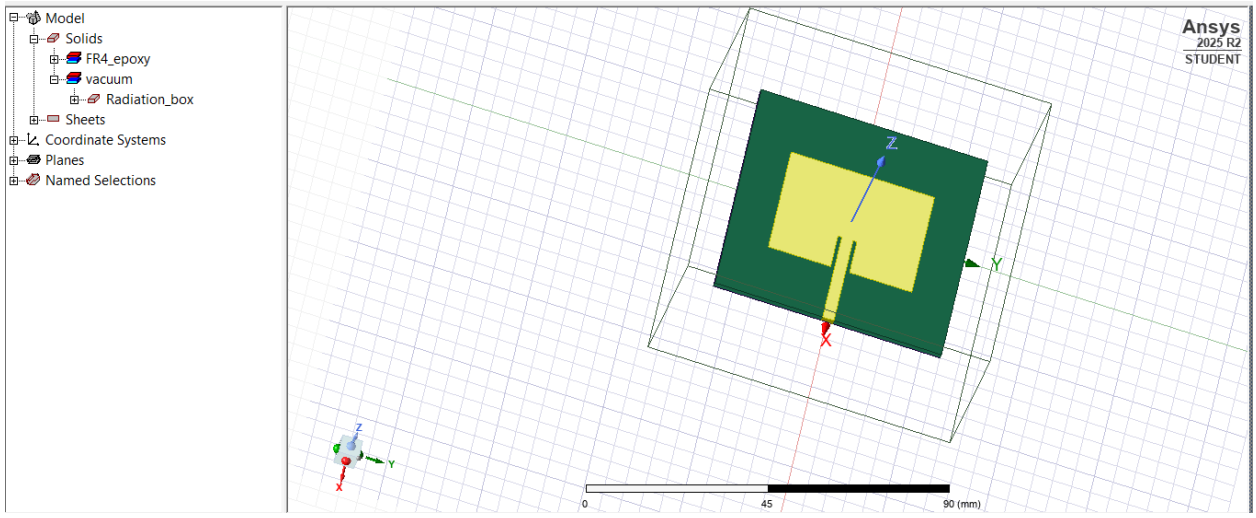

Fig 2

The analysis of the S-parameters indicated a return loss (S11) of around −20.8825 dB at the central frequency of 2.3820 GHz, as depicted in Figure 3. This low return loss reflects effective impedance matching at the feed point, resulting in minimal signal reflection. The measured bandwidth—defined as the frequency range over which S11 remains below −10 dB—was approximately 57.8 MHz. This exceeds the typical minimum requirement of 50 MHz for stable Wi-Fi performance, confirming the antenna's suitability for wireless communication systems.

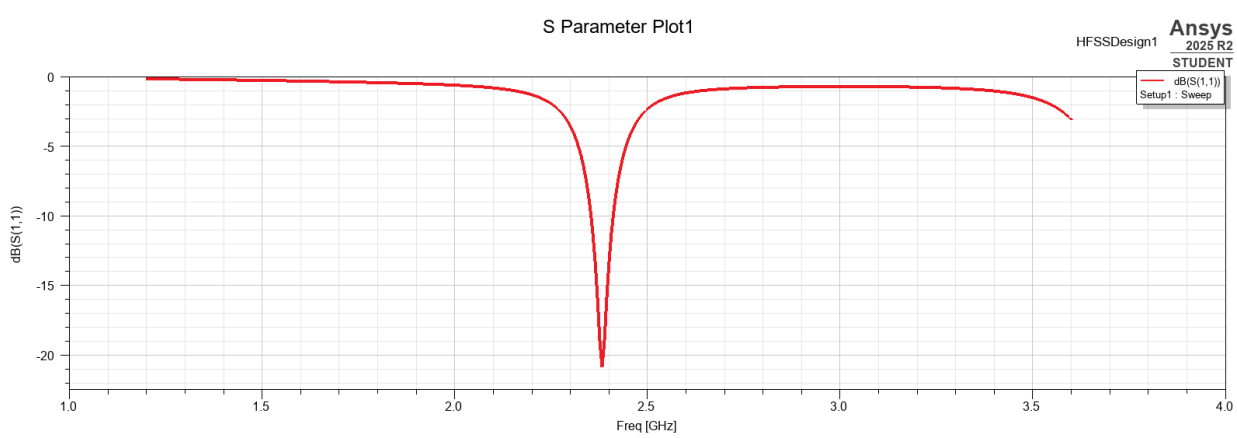

Fig 3

Figure 4 illustrates the 3D radiation pattern of the antenna at 2.4 GHz, offering a detailed representation of its radiative behavior. The antenna exhibited a directional gain of 10.56dB, with most of the radiated energy focused in the forward direction—typical of microstrip patch designs.

The smooth and symmetrical shape of the radiation pattern indicates consistent performance with low distortion, confirming stable and uniform radiation characteristics.

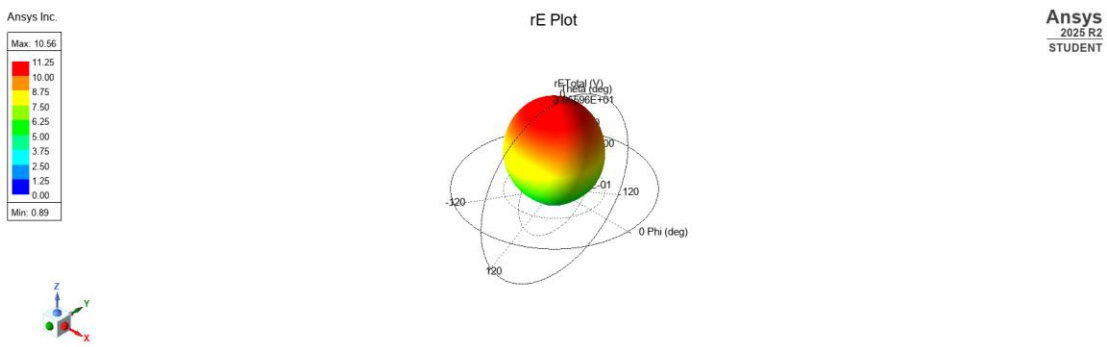

Fig 4

As illustrated in Figure 5, the surface current distribution at 2.4 GHz revealed high current concentration around the feed region and along the patch edges. This pattern indicates effective transfer of input power to the radiating element. The highest current density was observed near the feed point, validating efficient excitation and optimal radiation performance at the target frequency

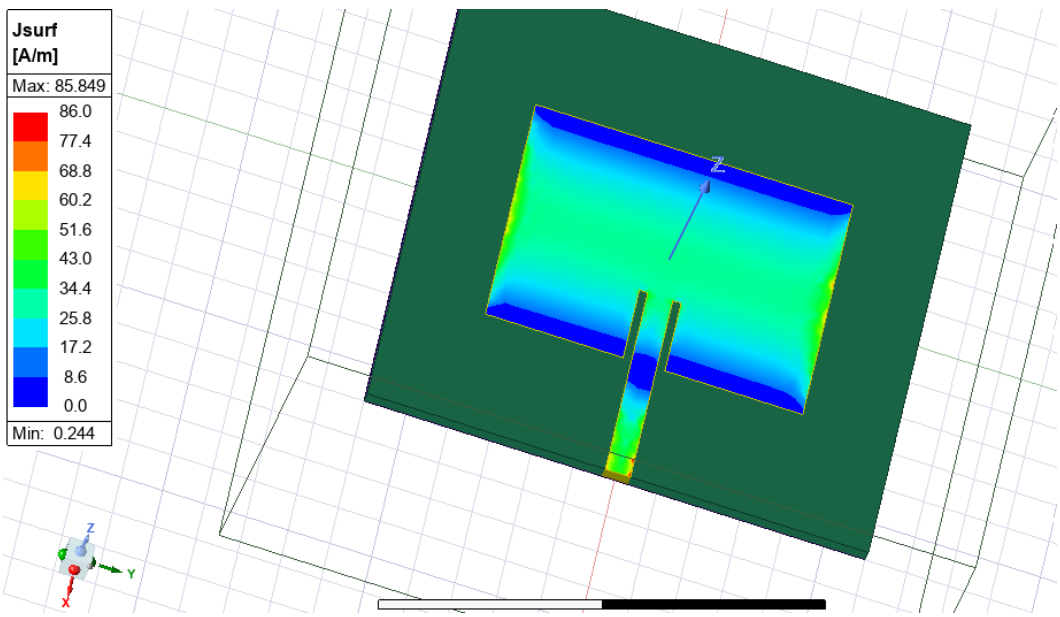

Fig 5

The Voltage Standing Wave Ratio plot design confirms that the S parameter simulation is correct and that that the simulation was highly successful. The most prominent feature of the graph is the deep "V" shaped dip, which indicates the antenna's resonance. This dip occurs almost exactly at 2.3820 GHz, proving that the antenna is tuned very close to the target frequency of 2.4 GHz. At this center frequency, the VSWR value drops to approximately 1.25, which is very close to the theoretical ideal of 1.0. This signifies a nearly perfect impedance match, also we can double check

our bandwidth calculations and see that its approximately 66 GHz which is very close to the value we got from the S parameter Plot.

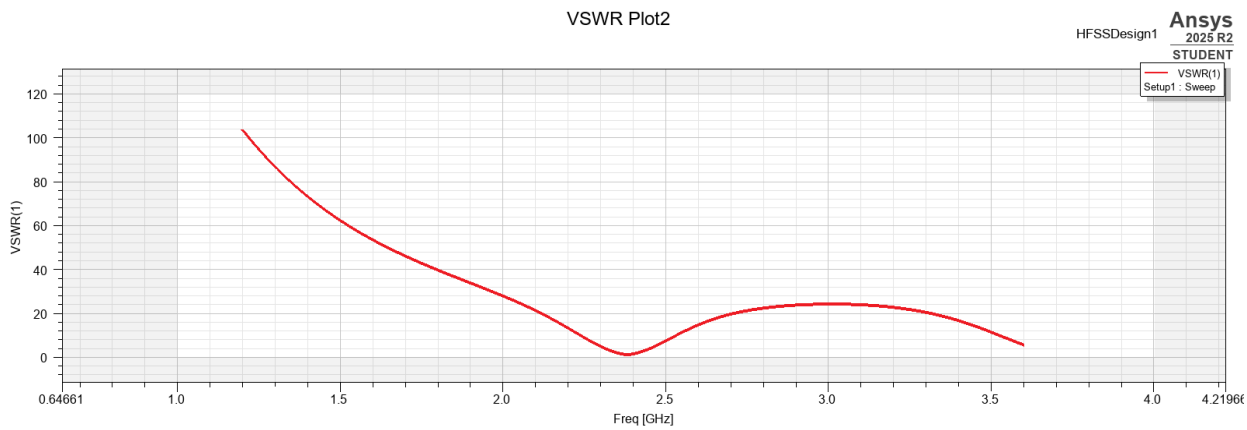

Fig 6

3. *Conclusions*:

The results of this model demonstrate the high utility of the rectangular microstrip patch antenna in directional communication. Through careful impedance matching, the design achieved a substantial gain of 11 dB. The resulting radiation pattern shows a marked increase in power density directed along the z-coordinate, validating the design's performance and suitability for high-gain applications.

4. *References*:

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